



AlphaServer ES45 Systems **Technical Summary**



Contents

System Overview 1

Features and Benefits 2

Third-Generation Alpha Chip 2

Chip Operation

Alpha 21264 Features

Model Variants 3

System Packaging 3

Chassis, Pedestal View

Chassis, Tower View

Rackmount Chassis

Architecture 4

System Block Diagram

System Board 5

Component and Connector Locations

Processor Module 6

Processor Configuration Rules

Memory 6

Memory Options

Memory Configuration

Memory Motherboard

Memory Performance Considerations

Memory Arrays and Sets on Memory Motherboards

System I/O 8

I/O Implementation

Bandwidth

I/O Ports

Block Diagram of I/O Control

System Control 10

Storage 10

Fibre Channel

RAID (Redundant Array of Independent Disks)

Server Management 10

Operational Management

Platform Management

Error Reporting

Security 11

Reliability and Availability Features 11

Processor Features

Memory Features

I/O Features

System Features

Installation and Maintenance 12

Clustering 12

PCI to Memory Channel Interconnect

Operating System Support

Performance and Benchmarking 13

Sources of Performance Information

Information for Compaq Partners

Service and Support 13

Hardware Warranty

Software Warranty

AlphaServer ES45 System Diagrams 14

System Features at a Glance 15

Physical Characteristics 16

Electrical Characteristics 16

HP AlphaServer ES45 Systems

The *HP AlphaServer ES45* system is a high-performance, scalable enterprise server for business, technical, and scientific applications. With its flexible design, it can go in an office environment or computer room. It performs as a high-capacity database server, high-performance application server, Network File System (NFS) server, or Internet server.

The *AlphaServer* products use the 64-bit Alpha RISC architecture that supports multiple operating systems:

Tru64 UNIX, *OpenVMS*, and Linux operating systems. The *AlphaServer ES45* system integrates into your current operating environment and anticipates future needs with upgrade capabilities.

For more information on *AlphaServer ES45* systems, see: <http://www.compaq.com/alphaserver>.

System Overview

The *AlphaServer ES45*, like its predecessor the *ES40*, uses the third-generation Alpha 21264 processor. The chip used in this system (21264C) is the EV68 running at 1000 MHz or at 1250 MHz. The switch-based system interconnect exploits the full potential of the Alpha 21264 chip.

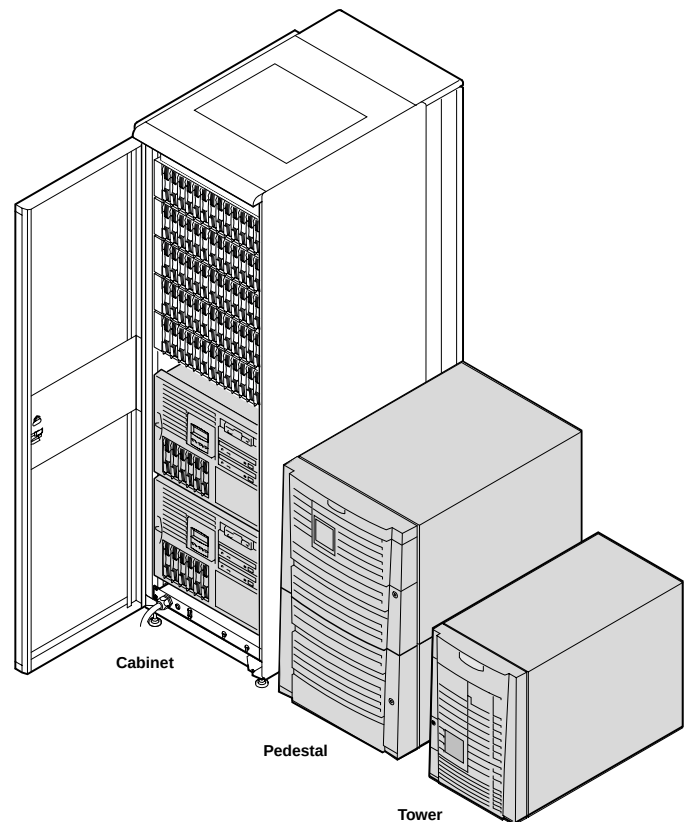
ES45 systems support up to four processors, 32 Gbytes of memory, and 10 PCI slots or 6 PCI slots and one 4X AGP slot. New features to the ES45 are a faster system bus (125 MHz), 66 MHz PCI slots with support for hot-plug PCI, and new high-bandwidth memory DIMMs.

Three model variants allow you to choose the PCI backplane that meets your needs.

The system is available in three packages: a standalone tower, a pedestal system with expanded storage capacity, and a rack-mount system.

The chassis used in the tower is rotated 90 degrees and installed in a pedestal enclosure or in a cabinet. Each chassis provides space for up to twelve 1-inch Ultra3 disks. Up to two Ultra3 Universal StorageWorks shelves can be installed in the pedestal system; and up to ten shelves (single-bus or split-bus) can be installed in a cabinet.

Systems can be purchased with *Tru64 UNIX* or *OpenVMS* operating systems installed. *Tru64 UNIX* systems can be purchased to allow customers to install Linux.



Features and Benefits

- **Leadership 64-Bit Architecture**

The Alpha 64-bit architecture was introduced with the Alpha 21064 chip in 1992 and now the 21264 builds upon that proven architecture.

- **Performance**

The Alpha 21264 chip, the world's fastest microprocessor, is offered with a switch-based interconnect that supports four processors. This switch-based system provides a memory bandwidth of up to 8 Gbytes/sec (peak) using two 256-bit memory buses running at 125 MHz.

- **Scalability**

AlphaServer ES45 systems can have up to four processors, up to 32 GB memory, and 10 I/O slots. The chassis can have two disk drive cages, with six disks each. In addition, the pedestal system provides space for another 28 disks beyond the 12 in the chassis. A cabinet can house up to five systems or up to ten StorageWorks shelves and one system.

- **Reliability and Availability**

The *AlphaServer* ES45 uses the latest technologies to achieve redundancy, error correction, and fault management. The system has redundant fans and power supplies; fans, power supplies, disks, and PCI options can be hot plugged. The remote management console (RMC) monitors, sends alerts, and records possible error conditions. The RMC can be accessed even if the system is completely down.

Third-Generation Alpha Chip

The third generation of the Alpha microprocessor, the Alpha 21264C, is a superscalar superpipelined implementation of the Alpha architecture. The first offering of this chip was known as EV6. These systems are offered with the EV68 chip (.18 micron). Each chip has over 15.2 million transistors.

Designed for performance, the 21264 achieves this goal by carefully studied and simulated architectural and circuit analysis. The 21264 memory system also enables the high performance levels. On-chip and off-chip caches provide for very low latency data access, which allows for very high bandwidth data access. (In ES45 systems with the 1000 MHz chip, the size of the off-chip cache is 8 MB running at 250 MHz DDR. With the 1250 MHz chip the size of the off-chip cache is 16 MB running at 313 MHz DDR.)

Internal to each chip is a 64-Kbyte instruction cache (I-cache) and a 64-Kbyte data cache (D-cache).

- **I-cache.** 64 Kbytes, two-way set-associative, virtually addressed cache with 64-byte blocks
- **D-cache.** 64 Kbytes, two-way set-associative, virtually indexed, physically tagged, writeback cache with 64-byte blocks

Chip Operation

Several key design choices were made in the chip architecture to maximize performance: Four instructions are fetched each cycle, and then the handling of those instructions boosts the speed of execution. Register renaming assigns a unique storage location with each write reference to a register, avoiding register dependencies that can be a potential bottleneck to processor performance.

Another design feature, out-of-order execution, permits instructions to execute in an order different from the order that the instructions are fetched. In effect, instructions execute as quickly as possible. This allows for faster execution, since critical path computations are started and completed as soon as possible.

In addition, the Alpha 21264 employs speculative execution to maximize performance. It speculatively fetches and executes instructions even though it may not know immediately whether the instruction will be on the final execution path. This is particularly useful, for instance, when the 21264 predicts branch directions and speculatively executes down the predicted path. The sophisticated branch prediction in the 21264 coupled with the speculative and dynamic execution extracts maximum instruction parallelism from applications.

For more information about the chip, see:

<http://www.compaq.com/alphaserver/download/ev6chip.pdf>

Alpha 21264 Features

- Out-of-order instruction execution
- 64 Kbyte on-chip data and instruction caches
- Improved branch prediction through intuitive execution
- Register renaming
- Increased bandwidth for high-speed access to second-level cache and system memory
- Motion video instructions
- Square root and divide instructions
- All instructions are 32 bits long and have a regular instruction format
- Floating-point unit, supports DIGITAL and IEEE floating-point data types
- 80 integer registers, 64 bits wide
- 72 floating-point registers, 64 bits wide

Model Variants

Model 1B (1 GHz) or Model 1 (1.25 GHz)

1–4 CPUs

Up to 16 GB memory

6 PCI slots – 2 at 33 MHz & 4 at 66 MHz

1 4X AGP slot

Model 2B (1 GHz) or Model 2 (1.25 GHz)

1–4 CPUs

Up to 32 GB memory

10 PCI slots – 4 at 33 MHz & 6 at 66 MHz

Model 3B (1 GHz) or Model 3 (1.25 GHz)

1–4 CPUs

Up to 32 GB memory

10 PCI slots – 8 at 33 MHz & 2 at 66 MHz

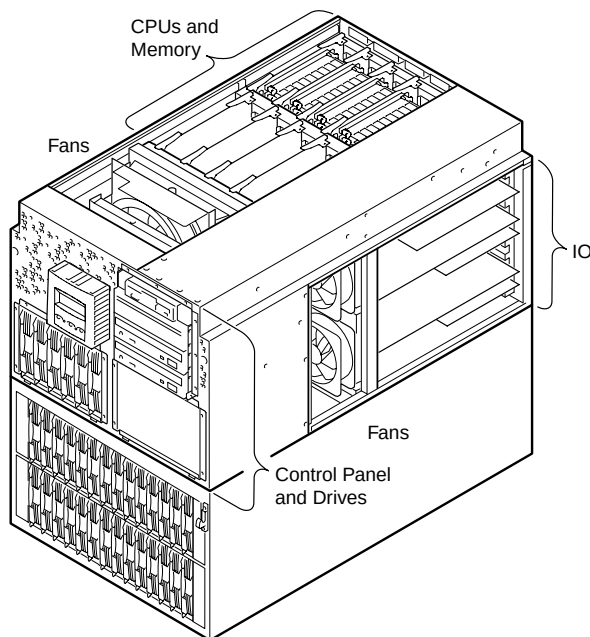
System Packaging

All system variants use the same chassis, with space for two disk cages in the front. Each cage holds up to six 1-inch Ultra3 SCSI disks.

The chassis has a compartment for the CPUs and memory and another one for the PCI I/O cards. The control panel, DVD-ROM and floppy drives, and two additional removable media bays are in the front, along with space for up to two disk cages. The power supplies are in the rear.

Chassis, Pedestal View

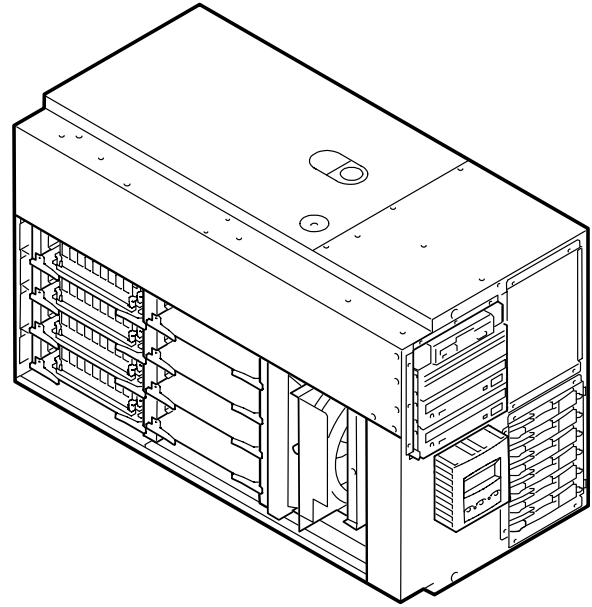
Additional storage is available in the pedestal enclosure. The chassis is mounted over a box that can house up to two Ultra3 Universal (single-bus or split-bus) StorageWorks shelves.



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Chassis, Tower View

In the tower enclosure, the chassis is rotated 90 degrees, with the control panel and removable media bays keeping their same horizontal position as in the pedestal/rack system.



MR0077

Rackmount Chassis

When installed in a cabinet, the chassis has the same orientation as that used in the pedestal. The chassis can be mounted in the 67-inch or 79-inch M-series cabinet, which is a 19-inch wide RETMA cabinet. Each system requires 14 inches (8U) of vertical space. Up to five systems can be mounted in the H9A15 79-inch cabinet and up to four systems in the H9A10 67-inch cabinet. If the maximum number of disks is desired, you can have ten StorageWorks shelves and one ES45 system in the taller cabinet, or one system and eight shelves in the other cabinet.

Architecture

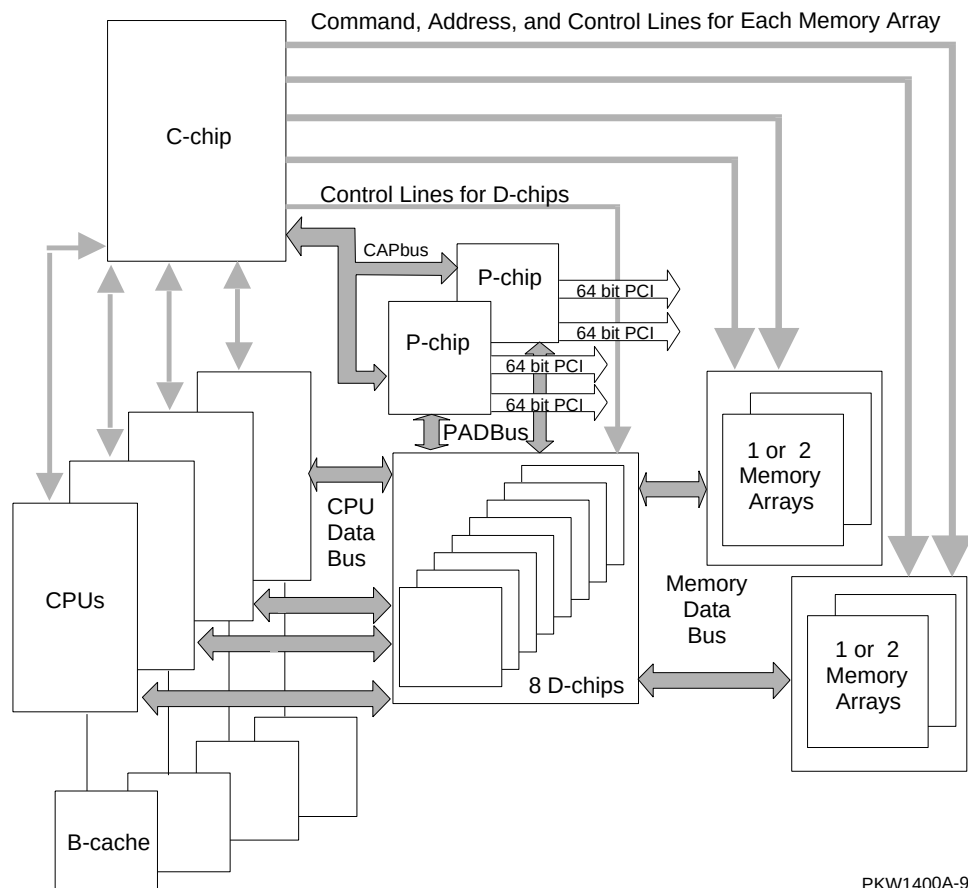
This system is designed to maximize the potential of the Alpha 21264 chip. The traditional bus interconnect has been replaced by a switch-based interconnect system. With a bus design, the processors, memory, and I/O modules share the bus. As the number of bus users increases, the transactions interfere with one another, increasing latency and decreasing aggregate bandwidth. However, with a switch-based system there is no degradation in performance as the number of CPUs, memory, and I/O users increase. Although the users increase, the speed is maintained.

With a switch-based, or point-to-point interconnect, the performance remains constant, even though the number of transactions multiplies. The switched system interconnect uses a set of complex chips that route the traffic over multiple paths. The chipset consists of one C-chip, two P-chips, and eight D-chips.

- **C-chip.** Provides the command interface from the CPUs and main memory. Although there is only one C-chip in the system, it allows each CPU to do transactions simultaneously.
- **D-chips.** Provide the data path for the CPUs, main memory, and I/O.
- **P-chips.** Provide the interface to four independent 64-bit PCI buses (Models 2B and 3B) or three 64-bit PCI buses and one 4X AGP bus. (Model 1B).

Two 256-bit memory buses support four memory arrays, yielding an 8 Gbyte/sec system bandwidth. Transactions are ECC protected. Upon the receipt of data, the receiver checks for data integrity and corrects any errors.

System Block Diagram



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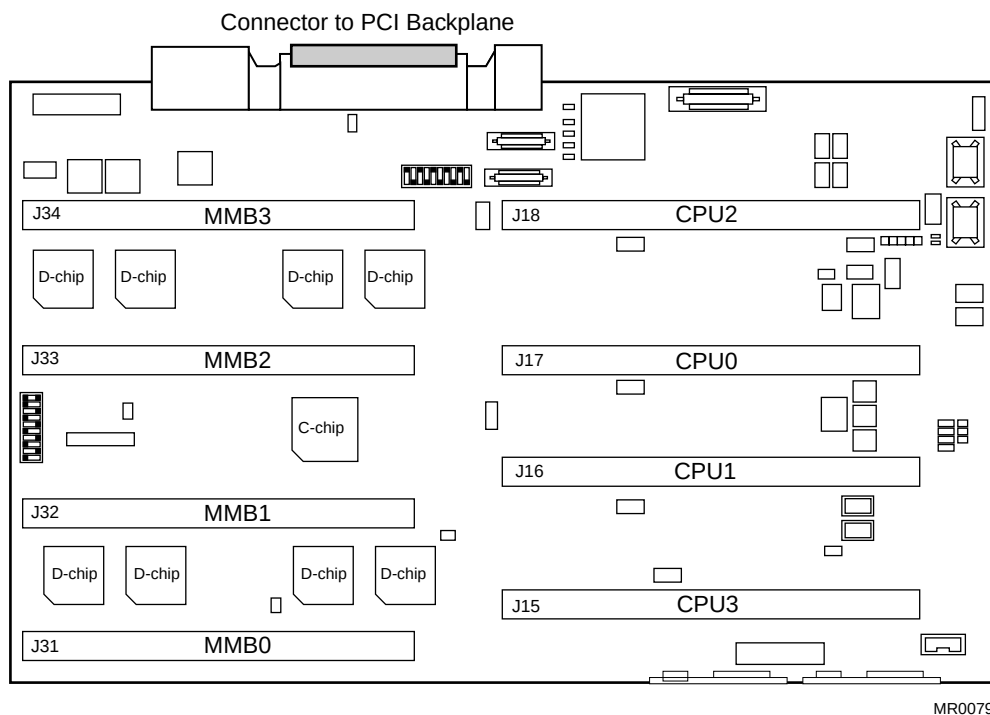
Model 1B has three PCI buses and one 4X AGP bus.

System Board

The entire chipset for the system interconnect (one C-chip, two P-chips, and eight D-chips) was implemented on the system board in the ES40. The figure below shows the system board for the ES45. In the ES45 the C-chip and D-chips are on the system board; the P-chips are on the PCI backplane. The chipset provides the data and address path between the CPUs, memory, and the I/O subsystem.

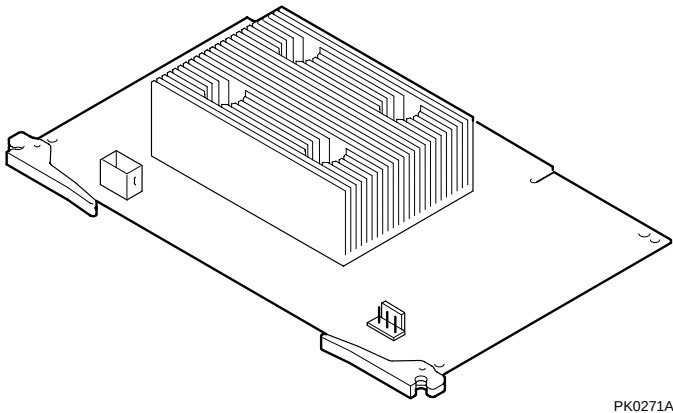
The system board has connectors for four CPU modules and four memory motherboards. On the back side of the module are connectors for the power supplies. The I/O connector provides both signals and power to the PCI backplane.

Component and Connector Locations



Processor Module

An AlphaServer ES45 can have up to four CPU modules. In addition to the Alpha chip, the CPU module has an 8-Mbyte (1 GHz) or 16-Mbyte (1.25 GHz) second-level cache and a DC-to-DC converter that provides the required voltage to the Alpha chip.



Processor Configuration Rules

- The first CPU module is installed in CPU slot 0.
- Additional CPUs are installed in the next available slot.

Memory

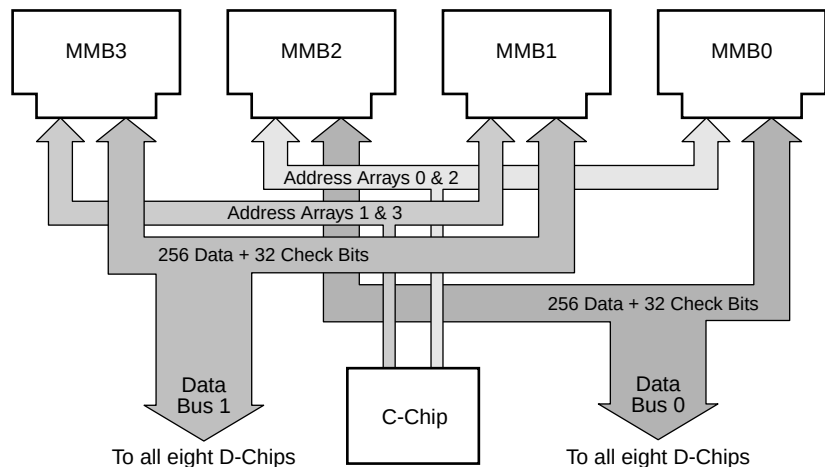
Memory throughput in this system is maximized by the following features:

- Two 256-bit wide memory data buses
- Very low memory latency (120 ns) and high bandwidth with 8 ns clock
- ECC memory

The switch interconnect can move a large amount of data over two independent memory data buses. Each data bus is 256 bits wide (32 bytes). The memory bus speed is 125 MHz. This yields 4 GB/sec bandwidth per bus ($32 \times 125 \text{ MHz} = 4 \text{ GB/sec}$). The maximum bandwidth is 8 GB/sec (4×2).

The design challenge was to maximize the capabilities of the two wide data buses. Distributing the 256 data bits equally over two memory motherboards (MMBs) was one solution: simultaneously, in a read operation, 128 bits come from one MMB and the other 128 bits come from another MMB, to make one 256-bit read. Another 256-bit read operation can occur at the same time on the other independent data bus.

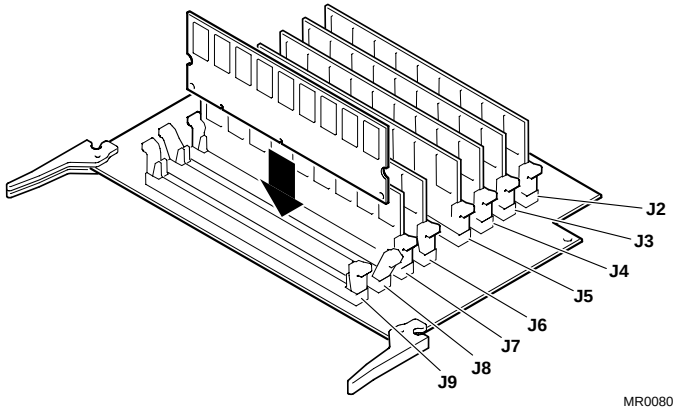
In addition, two address buses per MMB (one for each array) allow overlapping/pipelined accesses to maximize use of each data bus. When all arrays are identical (same size and speed), the memory is interleaved; that is, sequential blocks of memory are distributed across all four arrays.



Memory Options

The DIMMs in the ES45 system differ from those used in the ES40. With these DIMMs the memory bus runs at 125 MHz. The ES40 and ES45 DIMMs are **not** interchangeable. (The DIMMs can be plugged into the motherboard of the other system; however, a motherboard from one system physically cannot be installed into the other system.)

Memory Motherboard



Each memory option consists of four 200-pin industry-standard DIMMs with PECL clocks. The DIMMs are synchronous DRAMs. Memory options are available in the following sizes:

- 512 Mbytes (4 x 128 MB DIMMs)
- 1 Gbyte (4 x 256 MB DIMMs)
- 2 Gbytes (4 x 512 MB DIMMs)
- 4 Gbytes (4 x 1 GB DIMMs)

Each system has four memory motherboards. Models 2/2B and 3/3B accept up to eight DIMMs per motherboard, as shown in the artwork. Model 1/1B accepts four DIMMs per motherboard. Therefore, Models 2/2B and 3/3B support up to 32 Gbytes of memory while Model 1/1B supports up to 16 Gbytes.

Memory Configuration

A memory option (or set) consists of four DIMMs, and all four DIMMs must be the same size.

- Fill sets in numerical order. Populate all 4 slots in Set 0, then populate Set 1, and so on.
- An array is composed of two sets.
- DIMMs in an array must be the same size and type. For example, suppose you have populated Sets 0, 1, 2, and 3. When you populate Set 4, the DIMMs must be the same size and type as those installed in Set 0. Similarly, Set 5 must be populated with the same size and type of DIMMs as are in Set 1, and so on, as indicated in the following table.

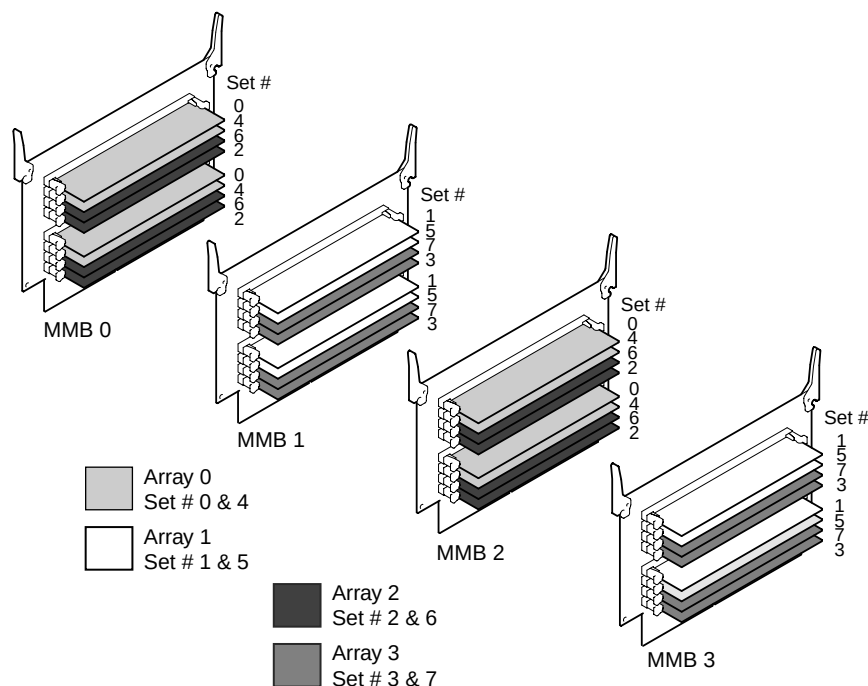
Array	System
0	Set 0 and Set 4
1	Set 1 and Set 5
2	Set 2 and Set 6
3	Set 3 and Set 7

- For optimum memory utilization and performance, memory DIMMs should be installed in the following order: 0, 1, 2, 3, 4, 6, 5, and 7.

Memory Performance Considerations

With one memory option (4 DIMMs) installed, memory operation interleaving will not occur. With two memory options (8 DIMMs), memory read-write operations are two-way interleaved. Interleaved operations reduce the average latency and increase the memory throughput over noninterleaved operations. With four memory options (16 DIMMs) installed, memory read-write operations are four-way interleaved, maximizing memory throughput.

Memory Arrays and Sets on Memory Motherboards



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System I/O

The industry-standard PCI bus is the number one choice for high-performance I/O options, such as disk storage and high-performance video applications.

I/O Implementation

The PCI bus implementation has the following characteristics:

- Fully compliant with the PCI Version 2.1 Specification
- Supports three address spaces: PCI I/O, PCI memory, and PCI configuration space
- Supports byte/word, tri-byte, and longword operations
- Exists in noncached address space only

Models 2/2B and 3/3B have four 64-bit PCI buses. Model 1/1B has three 64-bit PCI buses and a 4X AGP bus. All systems have hot-pluggable slots.

Model 2B (1 GHz) or Model 2 (1.25 GHz)

PCI Slot	PCI Bus Speed (MHz)	Signal Voltage	PCI Hot-Plug	PCI Hose Number
1	66	3.3	No	2
2	66	3.3	No	2
3	33	5.0	No	0
4	66	3.3	Yes	3
5	66	3.3	Yes	3
6	33	5.0	Yes	0
7	66	3.3	Yes	1
8	66	3.3	Yes	1
9	33	5.0	Yes	0
10	33	5.0	Yes	0

Model 1B (1 GHz) or Model 1 (1.25 GHz)

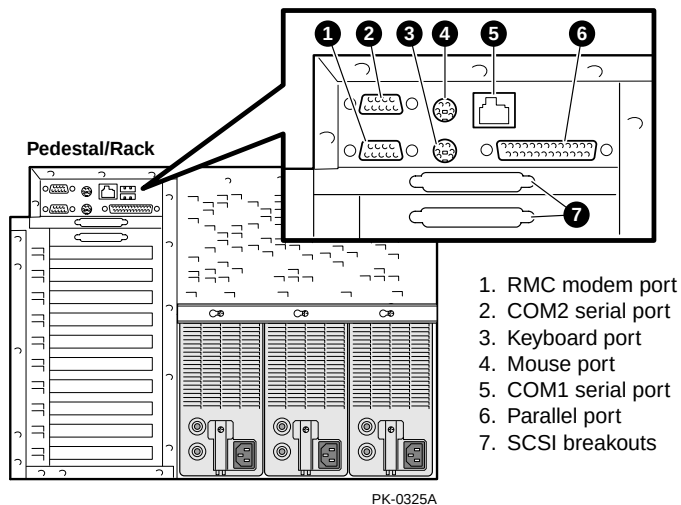
PCI Slot	PCI/AGP Bus Speed (MHz)	Signal Voltage	PCI Hot-Plug	PCI Hose Number
1	33	5.0	No	0
2	33	5.0	No	0
3	4X AGP	1.5	No	2
4	66	3.3	Yes	3
5	66	3.3	Yes	3
6				
7	66	3.3	Yes	1
8	66	3.3	Yes	1

Model 3B (1 GHz) or Model 3 (1.25 GHz)

PCI Slot	PCI Bus Speed (MHz)	Signal Voltage	PCI Hot-Plug	PCI Hose Number
1	33	5.0	No	2
2	33	5.0	No	2
3	33	5.0	No	0
4	66	3.3	Yes	3
5	66	3.3	Yes	3
6	33	5.0	Yes	0
7	33	5.0	Yes	1
8	33	5.0	Yes	1
9	33	5.0	Yes	0
10	33	5.0	Yes	0

I/O Ports

At the rear of the system are connectors offering access to two serial communications ports, one modem port for the remote management console, one parallel port, and ports for the keyboard and mouse. Two SCSI breakouts are also on the back panel.



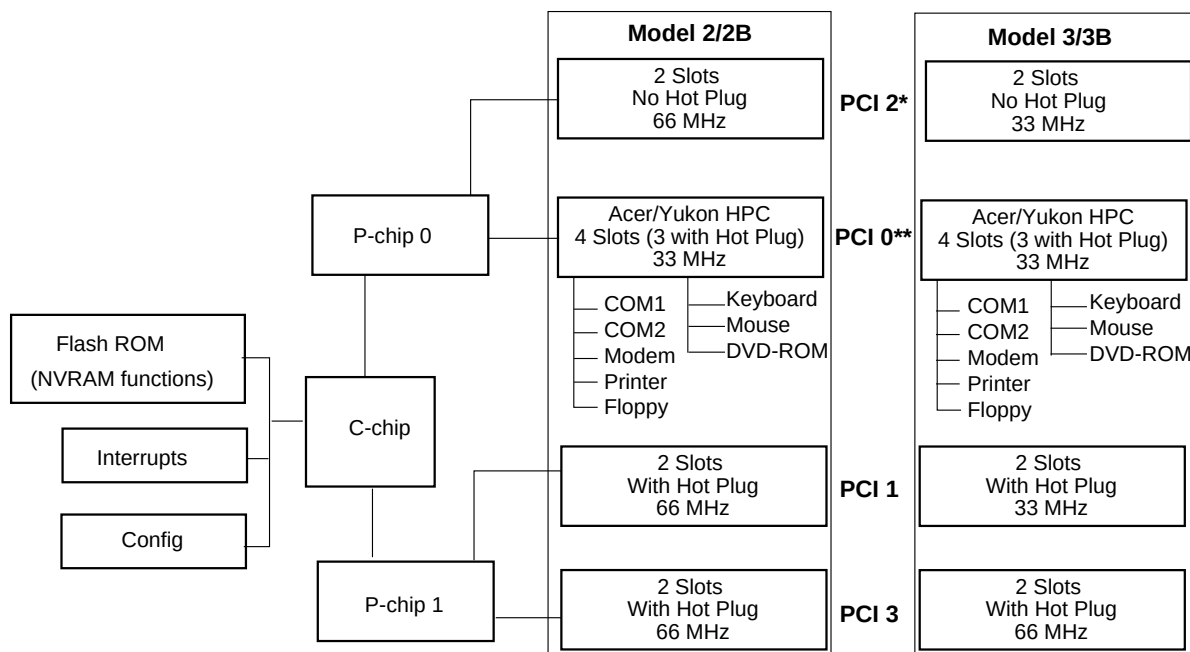
Bandwidth

The total PCI throughput for Model 2/2B (10 PCI slots) is 1.85 Gbytes/sec, and for Model 3/3B (also 10 PCI slots) is 1.32 Gbytes/sec. For Model 1/1B with six slots the PCI throughput is 1.32 Gbytes/sec. and the AGP throughput is 1 Gbyte/sec.

As can be seen in the block diagram, each P-chip controls two PCI buses, except in Model 1/1B where the AGP bus replaces one of the PCI buses. The Acer Labs 1543C chip provides the

bridge from PCI 0 to ISA. The C-chip controls accesses to memory on behalf of both P-chips.

Block Diagram of I/O Control



Model 1/1B * Has a 4X AGP bus with one slot in place of PCI 2.

** PCI 0 has only two slots - no hot plug.

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System Control

Close monitoring and control of the system environment and hardware is done by the remote management console (RMC). This logic also allows the system operator to monitor the system from a remote location. The RMC logic is implemented using a PIC17C44 microprocessor on the system board. The RCM firmware code resides on the microprocessor and in flash memory. The RMC is powered by an auxiliary 5V supply, so even when the system is powered off at the control panel the RMC can be accessed—so long as the system is plugged in.

The RMC provides the following monitoring and control functions:

- Monitors thermal sensors on the CPUs, the PCI backplane, and the power supplies
- Monitors voltages, power supplies, and fans
- Handles hot swapping of power supplies and fans
- Controls the control panel display and writes status messages on the display
- Detects alert conditions such as excessive temperature, fan failure, and power supply failure and sends an alert
- Shuts down the system if any fatal conditions exist
- Records error log information in nonvolatile RAM on each failing device

Storage

Removable media in the chassis includes a DVD-ROM drive, floppy diskette drive, and two 5.25 inch half-height bays, which can be combined for one 5.25 inch full-height bay.

A disk cage in the front of the chassis supports up to six Ultra3 SCSI Universal 1-inch disk drives (18.2, 36.4, or 72.8 GB). The drives are available at 10,000 or 15,000 RPM. Another disk cage can be added so that up to twelve disks can be installed. Each cage requires its own SCSI adapter.

The Ultra3 Universal StorageWorks shelf holds up to 14 disks. The pedestal system can include:

- Up to two Ultra3 Universal StorageWorks shelves (single-bus or split-bus)

The H9A15 cabinet can include:

- Up to ten Ultra3 Universal StorageWorks shelves

And the H9A10 cabinet can include:

- Up to eight Ultra3 Universal StorageWorks shelves

Fibre Channel

The ES45 supports Fibre Channel, which eliminates issues with today's SCSI interfaces such as distance, bandwidth, scalability, and reliability. Fibre Channel (FC) is the answer to not only server-to-storage connections but also to server-to-server networking, because multiple protocols are supported. SCSI, TCP/IP, video, or raw data can all take advantage of high-performance, reliable Fibre Channel technology.

With the KGPSA PCI Fibre Channel adapter, the ES45 systems provide a storage interconnect that is 2.5 times as fast as UltraSCSI: 100 vs 40 Mbytes/sec data throughput. The KGPSA adapter allows you to manage storage including the HSG80 RAID controller in a switched FC topology.

RAID (Redundant Array of Independent Disks)

The system can be configured with optional PCI RAID controllers to organize disk data cost-effectively, improve performance, and provide high levels of storage integrity.

The optional RAID controllers have the following features:

- Support for hot-swap drives
- Automatic rebuild after hot swap
- Console support for booting system from RAID
- RAID levels 0, 1, 0+1, 5
- Optional write cache
- Optional read cache
- Support for command queuing

Server Management

The *AlphaServer* products support important operational and platform management requirements.

Operational Management

Server/Network Management. *Compaq Insight Manager* is included with every system. This software tool allows you to monitor and control Alpha based servers. *Insight Manager* consists of two components: a Windows-based console application and server- or client-based management data collection agents. Management agents monitor over 1,000 management parameters. Key subsystems are instrumented to make health, configuration, and performance data available to the agent software. The agents act upon that data, by initiating alarms in the event of faults and by providing updated management information, such as network interface or storage subsystem performance statistics.

Remote Server Management. The integrated remote management console (RMC) lets the operator perform several tasks from a serial console: monitor the system power, temperature, and fans, and reset, halt, and power the system on or off. The monitoring can be done locally or remotely through a modem.

Platform Management

The *AlphaServer ES45* systems support platform management tasks such as manipulating and monitoring hardware performance, configuration, and errors. For example, the operating systems provide a number of tools to characterize system performance and display errors logged in the system error log file.

In addition, system console firmware provides hardware configuration tools and diagnostics to facilitate quick hardware installation and troubleshooting. The system operator can use simple console commands to show the system configuration, devices, boot and operational flags, and recorded errors. Also, the console provides inventory support and configuration management by giving access to serial numbers and revisions of hardware and firmware.

Error Reporting

Compaq Analyze, a diagnostic service tool used to determine the cause of hardware failures, is installed with the operating systems. It provides automatic background analysis, as it constantly views and reads the error log file. It analyzes both single error/fault events and multiple events. When an error condition is detected, it collects the error information and sends it and an analysis to the user. The tool requires a graphics monitor for its output display.

Security

- Front doors can be locked to prevent access to the disk drives and the rest of the system.
- An interlock sensor switch shuts down power if the top cover to the CPU/memory area is removed while power is on.
- Password protection is offered by the SRM console and RMC.

Reliability and Availability Features

The *AlphaServer ES45* system achieves an unparalleled level of reliability and availability through the careful application of technologies that balance redundancy, error correction, and fault management. Reliability and availability features are built into the CPU, memory, and I/O, and implemented at the system level.

Processor Features

- CPU data cache provides error correction code (ECC) protection.
- Parity protection on CPU cache tag store.
- Multi-tiered power-up diagnostics to verify the functionality of the hardware.

When you power up or reset the system, each CPU, in parallel, runs a set of diagnostic tests. If any tests fail, the failing CPU is configured out of the system. Responsibility for initializing memory and booting the console firmware is transferred to another CPU, and the boot process continues. This feature ensures that a system can still power up and boot the operating system in case of a CPU failure. Messages on the operator control panel power-up/diagnostic display indicate the test status and component failure information.

Memory Features

- The memory ECC scheme is designed to provide maximum protection for user data. The memory scheme corrects single-bit errors and detects double-bit errors and total DRAM failure.
- Memory failover. The power-up diagnostics are designed to provide the largest amount of usable memory, configuring around errors.

I/O Features

- ECC protection on the switch interconnect and parity protection on the PCI and SCSI buses.
- Extensive error correction built into disk drives.
- Optional internal RAID improves reliability and data security.
- Hot-plug disks.
- Hot-plug PCI options.

System Features

Auto reboot. On systems running *Tru64 UNIX* or *OpenVMS*, a firmware environment variable lets you set the default action the system takes on power-up, reset, or after an operating system crash. For maximum system availability, the variable can be set to cause the system to automatically reboot the operating system after most system failures.

Software installation. The operating systems are factory installed. Factory installed software (FIS) allows you to boot and use your system in a shorter time than if you install the software from a distribution kit.

Diagnostics. During the power-up process, diagnostics are run to achieve several goals:

- Provide a robust hardware platform for the operating system by ensuring that any faulty hardware does not participate in the operating system session. This maximizes system uptime by reducing the risk of system failure.
- Enable efficient, timely repair.

Audible beep codes report the status of diagnostic testing. The system has a firmware update utility (LFU) that provides update capability for console and PCI I/O adapter firmware. A fail-safe loader provides a means of reloading the console in the event of corrupted firmware.

Thermal management. The air temperature and fan operation are monitored to protect against overheating and possible hardware destruction. Six fans provide front to back cooling, and the power supplies, in the rear, have their own fans. If the temperature rises, the system fans speed up; or if necessary to prevent damage, the system shuts down. If the main fan, which cools the system card cage, fails, a redundant fan takes over. Fans can be replaced while the system is running (hot-plug).

Error handling. Parity and other error conditions are detected on the PCI buses. The memory checking scheme corrects single-bit errors and detects double-bit errors. Multiple ECC corrections to single-bit errors detected by the operating systems help in determining where in the system the error originated. Errors are logged for analysis.

Hot-plug disks. The hardware is designed to enable hot plug of disks. Disks can be removed and replaced while the rest of the system remains powered on and continues to operate. This feature contributes significantly to system availability. Since many disk problems can be fixed without shutting down the entire system, users lose access only to the disks that are removed.

Hot-plug PCI. PCI options can be replaced or new ones added while the system is running. In the 10-slot PCI box, seven slots are hot pluggable; four slots are hot pluggable in Model 1/1B. Systems are “hardware ready”; complete support to come.

N+1 power redundancy. A second or third power supply can be added to provide redundant power to the chassis. A second power supply is needed for more than two CPUs, more than one disk cage, and more than 16 DIMMs. In this case the third supply provides redundancy. Power supplies are 1080 watts (DC). Each has two LEDs to indicate the state of power to the system. In systems with redundant power supplies, a faulty power supply can be replaced while the system continues to run. Three AC power cords allow the system to be powered from three different sources, so that if one AC source fails, the system remains running.

An external UPS can be purchased to support critical customer configurations. Because power is maintained for the entire system (CPU, memory, and I/O), power interruptions are completely transparent to users.

Installation and Maintenance

The systems are designed for easy hardware, software, and option installation. Options ordered with a system are pre-installed and tested at the factory. The operating systems are also installed at the factory.

Additional CPUs, memory, power supplies, disks, and PCI options can be added to the tower and pedestal systems by anyone with appropriate technical training and experience. Installation of components in a rackmount system is reserved for service providers and self-maintenance customers.

Clustering

A cluster is a loosely coupled set of systems that behaves (is addressed and managed) like a single system, but provides high levels of availability through redundant CPUs, storage, and data paths. Clusters are also highly scalable; that is, CPU, I/O, storage, and application resources can be added incrementally to efficiently increase capacity. For customers, this translates to reliable access to system resources and data, and investment protection of both hardware and software.

Clustering allows multiple computer systems to communicate over a common interface, share disks, and spread the computing load across multiple CPUs. Clustering is implemented using our traditional interconnects and using the newest technology.

PCI to Memory Channel Interconnect

Under *Tru64 UNIX* and *OpenVMS*, you can build high-availability clusters using the PCI to Memory Channel interconnect. The Memory Channel interconnect is a high-bandwidth, low-latency PCI-based communications interconnect for up to eight *AlphaServer* systems. Data written to one computer's memory is shared by other computers on the Memory Channel bus.

The PCI adapter is the interface between a PCI and a Memory Channel bus. This bus is a memory-to-memory computer system interconnect that permits I/O space writes in one computing node to be replicated into the memories of all other nodes on the Memory Channel bus. A write performed by any CPU to its reflected address region results in automatic hardware updates to memory regions in other nodes. One node's write is “reflected” to other nodes as a direct side effect of the local write. This provides a memory region with properties similar to a high-performance shared memory across a group of nodes.

Operating System Support

For clustered *Tru64 UNIX* systems, TruCluster Software solutions allow users access to network services and provide further failover recovery from server, network, or I/O failures. *Tru64 UNIX* cluster systems use the SCSI bus and/or PCI to Memory Channel interconnect bus between disks and systems.

OpenVMS cluster systems use the CI, SCSI, Ethernet, FDDI, and Memory Channel as the interconnect between disks and systems.

The primary means of clustering *AlphaServer* ES45 systems depends on the operating system.

- CI clusters, *OpenVMS* only
- Memory Channel, *Tru64 UNIX* and *OpenVMS*
- SCSI clusters, *Tru64 UNIX* and *OpenVMS*

Performance and Benchmarking

HP has an ongoing program of performance engineering, using industry-standard benchmarks that allow comparisons across major vendors' systems. These benchmarks against competitive systems are based on comparable CPU performance, coupled with comparable memory and disk expandability.

See Table 1 for the record-breaking performance numbers of the *AlphaServer* ES45 systems.

System performance, however, is highly dependent upon application characteristics. Thus, benchmark information is one helpful "data point" to be used in conjunction with other purchase criteria such as features, service, and price.

Sources of Performance Information

Performance information is available on the Internet.

<http://www.compaq.com/alphaserver/performance/index.html>

http://www.ideasinternational.com/benchmark/spec/specfp_s2000.html

Information for Compaq Partners

If you are a Channel or Reseller Partner, you can find the tools, resources, and information you need to conduct business online on the secure Compaq Business Partner site:

<http://bps.compaq.com/businesspartner>

Also see the following:

Compaq Solutions Alliance site at <http://csa.compaq.com>

and HP's Developer and Solution Partner Portal (DSPP),

<http://www.hp.com/go/partners>

Service and Support

HP provides a comprehensive set of services that range from migration, consulting, and training, to direct support of Alpha systems, software, and applications. For information on services, see

<http://www.compaq.com/services>.

<http://www.hp.com/hps/>

Hardware Warranty

The *AlphaServer* ES45 system and components, including CPU, memory, PCI controllers, and power supplies, have a 1-year on-site, 5-day per week, 9-hour per day hardware warranty with next-day response time.

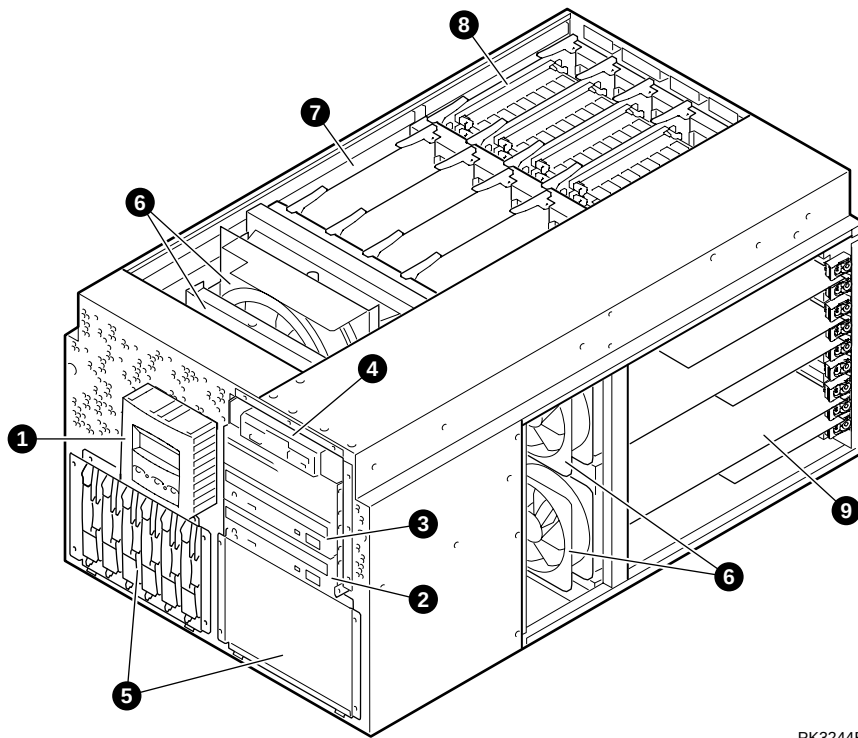
StorageWorks components contained in the pedestal or cabinet systems are supported by the comprehensive StorageWorks warranty: five years for disks, three years for controllers, two years for tape devices, and one year for other components. The first year includes on-site next-day response time. Network products in the pedestal or cabinet systems carry the network products warranty.

Users can upgrade to higher levels of service through a variety of hardware supplemental services.

Software Warranty

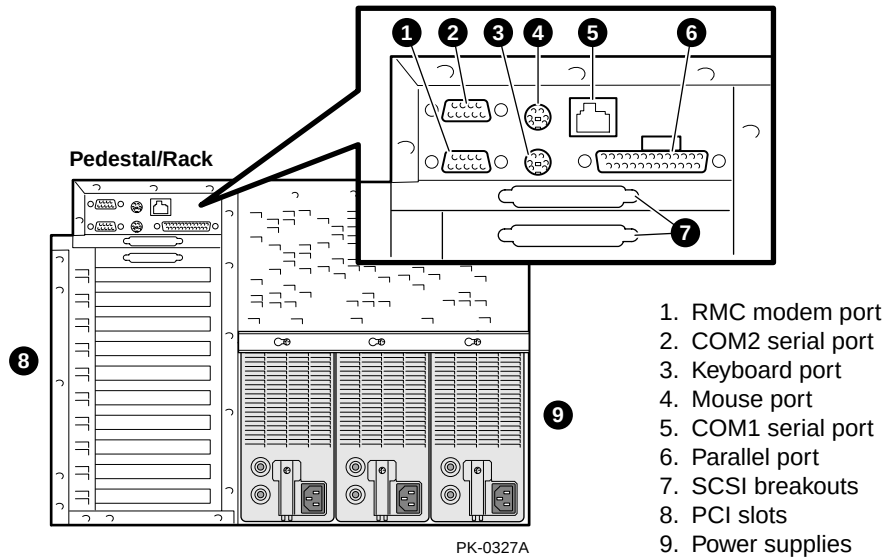
The warranty for *Tru64 UNIX* and *OpenVMS* is conformance to SPD with advisory telephone support for a period of 90 days. Users can upgrade to higher levels of service through a variety of software supplemental services.

AlphaServer ES45 System Diagrams



1. Control panel
2. DVD-ROM drive
3. Removable media bays
4. Floppy diskette drive
5. Storage drive bays
6. Fans
7. CPUs
8. Memory
9. PCI slots

PK3244B



1. RMC modem port
2. COM2 serial port
3. Keyboard port
4. Mouse port
5. COM1 serial port
6. Parallel port
7. SCSI breakouts
8. PCI slots
9. Power supplies

PK-0327A

System Features at a Glance

Table 1 provides a quick reference to features of the *HP AlphaServer ES45* systems.

Table 1 AlphaServer ES45 Features

CPU Features	68/1000			68/1250		
Processor	1–4 Alpha 21264C chips			1–4 Alpha 21264C chips		
CPU clock speed	1000 MHz			1250 MHz		
Cache on chip	64 KB I-cache			64 KB I-cache		
	64 KB D-cache			64 KB D-cache		
On-board cache	8 MB ECC DDR			16 MB ECC DDR		
Memory	Model 1B (1GHz)	Model 2B and 3B (1GHz)				
	Model 1 (1.25GHz)	Model 2 and 3 (1.25GHz)				
Capacity (maximum)	16 GB	32 GB				
Bandwidth (peak)	8 GB/sec	8 GB/sec				
Performance	68/1000			68/1250		
	1 CPU	2 CPUs	4 CPUs	1 CPU	2 CPUs	4 CPUs
SPECint2000	679	–	–	928	–	
SPECfp2000	960	–	–	1327	–	
SPECint_rate2000	7.87	15.6	30.8	10.80	21.3	42
SPECfp_rate2000	11.1	21.1	36.8	15.4	28.1	48.8
SPECjbb2000		22,126	42,830	–	–	–
TPC-C	–	–	50,117 tpmC @ \$15.24/tpmC *	–	–	56,375 tpmC @ \$9.39/tpmC **
Standard Features	1.44 MB diskette drive, DVD-ROM drive, 1 RMC dedicated serial port, 2 serial ports, 1 parallel port, integrated remote system management console, operating system license and customer documentation, Internet software					
Storage						
Removable media	DVD-ROM, floppy diskette, two HH 5.25” removable media bay					
System chassis	12 disks (873.6 GB based on 72.8 GB disks) Ultra3 SCSI					
Pedestal	Disk Drive Cages	PLUS	Universal StorageWorks Split-Bus Shelves			
	12 disks @ 72.8 GB = 873.6 GB		2 shelves: 28 disks @ 72.8 GB = 2038 GB			
	Total capacity: 2.9 TB					
H9A10 cabinet	Disk Drive Cages	PLUS	Universal StorageWorks Split-Bus Shelves			
	One system (2 cages, 6 per)		6 shelves (14 disks per shelf)			
	12 disks @ 72.8 GB = 873.6 GB		84 disks @ 72.8 GB = 6115.2 GB			
	Total capacity: 6.9 TB					
H9A15 cabinet	Disk Drive Cages	PLUS	Universal StorageWorks Split-Bus Shelves			
	One system (2 cages, 6 per)		10 shelves (14 disks per shelf)			
	12 disks @ 72.8 GB = 873.6 GB		140 disks @ 72.8 GB = 10,192 GB			
	Total capacity: 11.1 TB					
I/O System	Model 1B (1GHz)		Model 2B (1GHz)	Model 3B (1GHz)		
	Model 1 (1.25GHz)		Model 2 (1.25GHz)	Model 3 (1.25GHz)		
I/O slots	6 PCI slots	1 4X AGP slot	10 PCI slots	10 PCI slots		
Maximum throughput	1.32 GB/sec	1 GB/sec	1.85 GB/sec	1.32 GB/sec		
High Availability Features						
System	System auto reboot, thermal management, remote system management, RAID 0, 1, 0+1, 5, hot plug of disks, power supplies, fans, and PCI options, memory failover, ECC memory, ECC cache, N+1 power supply, SMP CPU failover, error logging, optional uninterruptible power supply					
OpenVMS clusters	CI, Ethernet, SCSI, FDDI, PCI to Memory Channel Interconnect					
Tru64 UNIX TruClusters Solutions	SCSI, PCI to Memory Channel Interconnect					
Operating Systems	Tru64 UNIX, OpenVMS, Linux					
Warranty						
Hardware	1-year, on-site, 5 day x 9 hour warranty with 24-hour response					
Software	90-day telephone advisory support for OpenVMS and Tru64 UNIX					

* Database is Oracle 91 using nine Compaq Smart Array 5300 backplane RAID controllers.

** Database is Sybase V12.5 using six Compaq Smart Array 5300 backplane RAID controllers.

Physical Characteristics

Table 2 details basic physical characteristics of the system, and Table 3 gives the electrical characteristics.

Table 2 AlphaServer ES45 Physical Characteristics

Dimensions	Tower	Pedestal	Rackmount
Height	50.8 cm (20.0 in.)	78.2 cm (30.8 in.)	35.2 cm (13.87 in.)
Width	38.7 cm (15.25 in.)	50.8 cm (20.0 in.)	44.7 cm (17.6 in.)
Depth	80.5cm (31.7 in.)	82.3 cm (32.4 in.)	78.2 cm (30.8 in.)
Weight	65 kg (143 lb) typical; 96 kg (211 lb) maximum	127 kg (280 lb) typical; 159 kg (350 lb) maximum	With brackets, slides, & cables Nominal 59 kg (130 lb) Maximum 92 kg (202.4 lb)

Environmental

Temperature	Operating Storage (60 days) Rate of change	10–35° C (50–95° F) –40 to 66° C (–40 to 151° F) 11° C/hr (20° F/hr)
Relative humidity	Operating Nonoperating Storage (60 days) Rate of change	10–90% 10–90% 10–95% 20%/hr.
Maximum wet bulb temperature	Operating Storage (60 days)	28° C (82° F) 46° C (115° F)
Minimum dew point temperature	Operating	2° C (36° F)
Heat dissipation	Nominal	Maximum
Tower and Rack	980 w, 3347 BTU/hr	1440 w, 4918 BTU/hr
Pedestal	1560 w, 5328 BTU/hr	2400 w, 8196 BTU/hr
H9A10/H9A15	Configuration dependent	4800 w, 16,392 BTU/hr
Airflow and quality	Intake location Exhaust location	Front Rear–Tower, Pedestal, Rack Rear/Top–H9A10/H9A15
Altitude	Operating Nonoperating	3,037 m (10,000 ft) 12,190 m (40,000 ft)
Vibration	Operating	10–500 Hz .1 G peak
Mechanical shock	Operating Tower/Pedestal M-Series Cabinet	7.5 G, 10 +/- 3ms 5.0 G, 10 +/- 3ms

Acoustics—Declared values per ISO 9296 and ISO 7779

Product	L _{WAd} ¹ B		L _{pAm} ¹ dBA (bystander positions)	
	Idle	Operate	Idle	Operate
Rackmount or Tower	6.4	6.4	47	47
Pedestal	6.6	6.6	48	48
StorageWorks Model 43xx	6.3	6.4	44	45

Current values for specific configurations are available from HP representatives. 1 B = 10 dBA.

Table 3 AlphaServer ES45 Electrical Characteristics

Nominal voltage (Vac)	200–240
Voltage range (Vac) temporary condition)	180–250
Power source phase	Single
Nominal frequency (Hz)	50/60
Frequency range (Hz)	47–53/57–63
RMS current (max. steady state)	
Tower and Rackmount	
Single power cord	5.6 A
Multiple power cords	3.6 A
Max VA	1440 VA
Pedestal	
Each power cord	6.0A
Max VA	2070 VA
M-Series cab config.-dependent	
Each power cord	16 A



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